



# Business Intelligence & MIS Analytics

## Customized Packaging Business

From Operational Data to Management Insights

### ORDER-TO-CASH FLOW



Built on a synthetic dataset created for this interview project. No figure in this deck represents actual company performance.

# Data sits in six functions — management needs one consolidated view

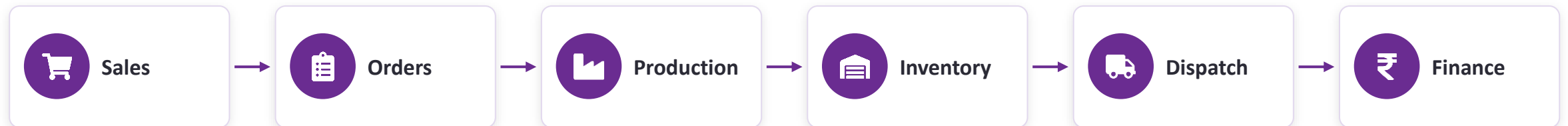
## WHAT THE COMPANY OFFERS



## WHY A CONSOLIDATED VIEW

-  Data is distributed across Sales, Customers, Production, Inventory, Dispatch and Finance.
-  Management needs consolidated, timely reporting — not separate functional extracts.
-  Revenue, margin, delivery and cash decisions depend on one version of the truth.

## END-TO-END BUSINESS PROCESS

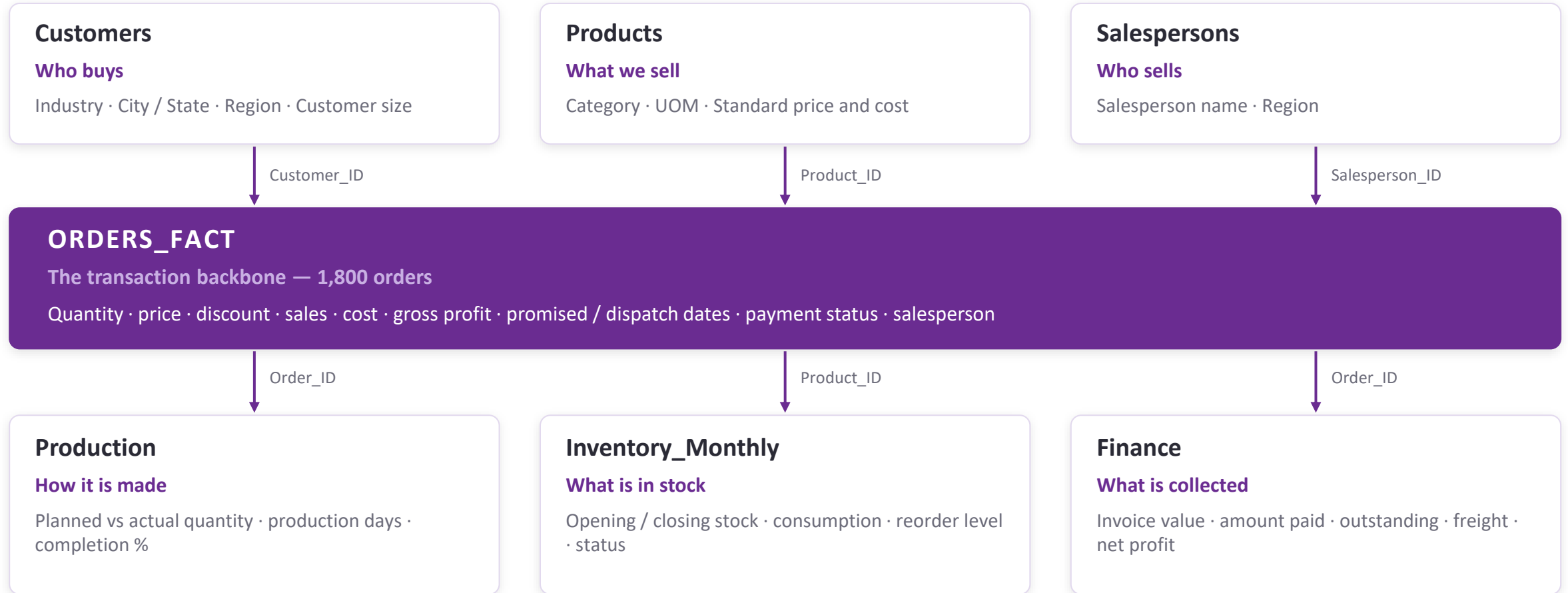


## PROBLEM STATEMENT

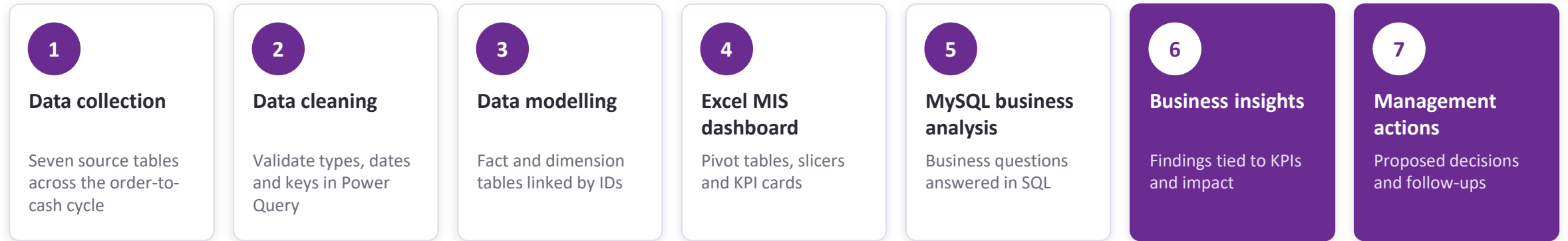
How can integrated business data be transformed into actionable insights for better revenue, profitability, operational efficiency and working-capital decisions?

# Seven tables, one order-centred data model

## MASTER DATA · WHO, WHAT AND BY WHOM



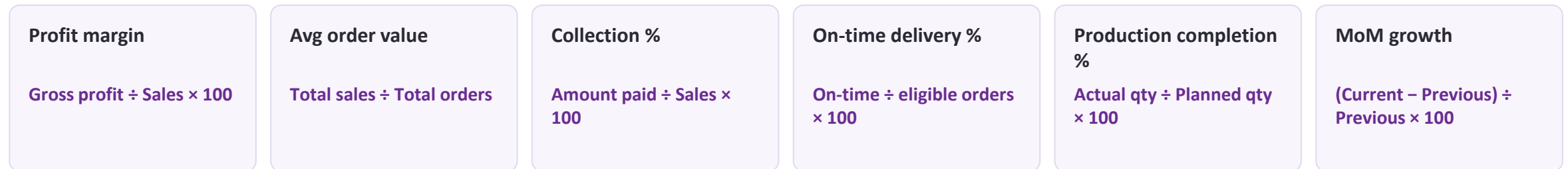
# A repeatable path from raw tables to management actions



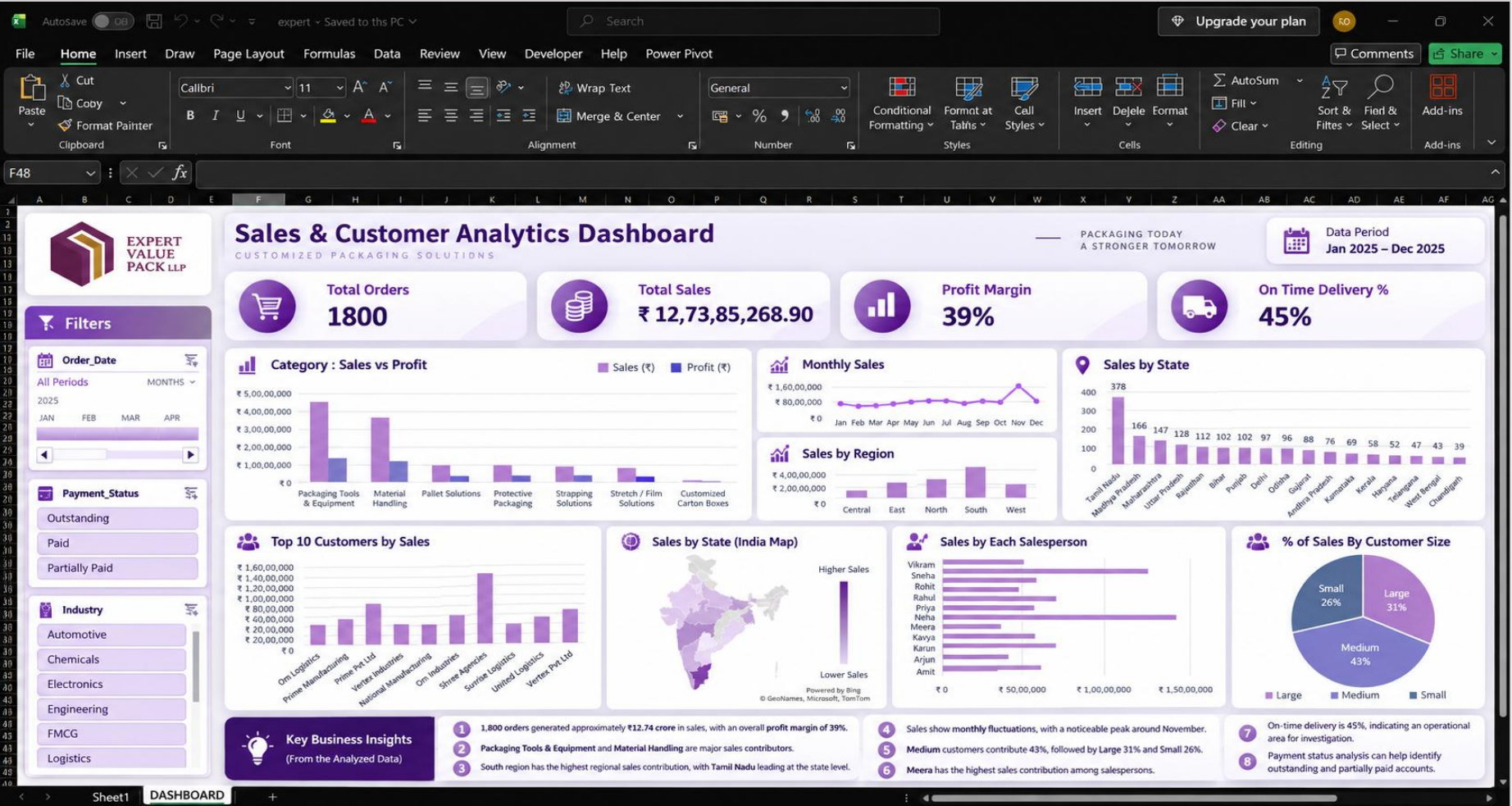
## TOOLKIT



## HOW THE BUSINESS KPIs ARE DEFINED



# One page gives management the state of the business



1,800  
Total orders

₹12.74 Cr  
Total sales

32.2%  
Gross margin (GP ÷ Sales)

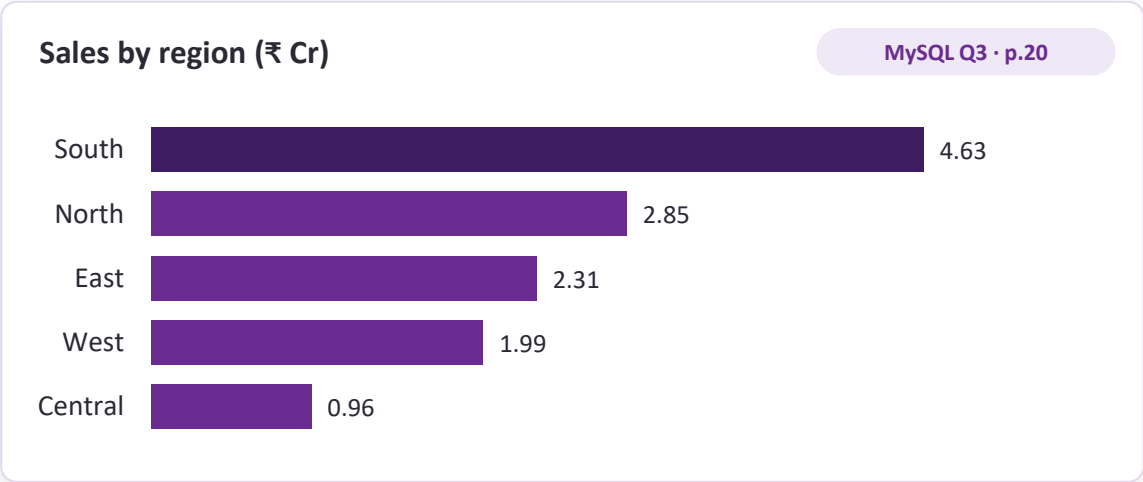
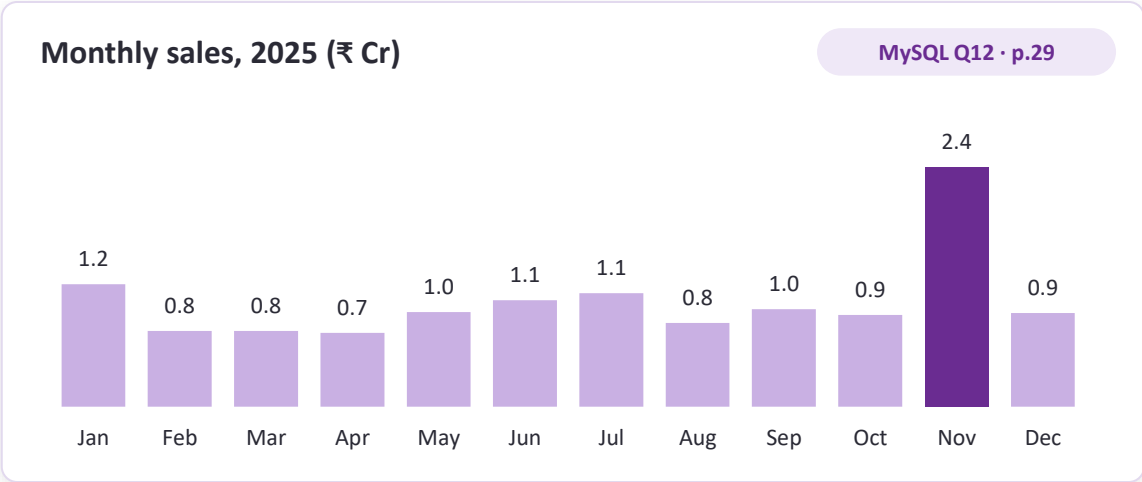
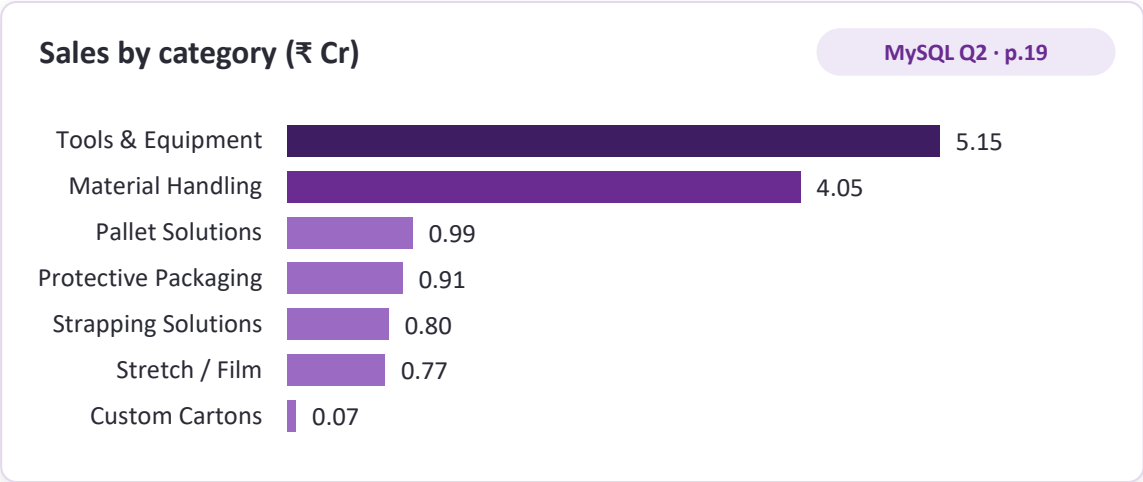
45%  
On-time delivery

Interactive slicers: Order Date · Payment Status · Industry | Period: Jan–Dec 2025 (synthetic)

Margin note: the Excel dashboard card shows 39%; this deck reports the SQL-weighted margin (gross profit ₹4.10 Cr ÷ sales ₹12.74 Cr = 32.2%) so every figure reconciles with the SQL appendix.

Synthetic dataset built for interview purposes — figures do not represent actual company performance

# Two categories and one region carry most of the synthetic revenue



**State-level and market concentration**

## Tamil Nadu

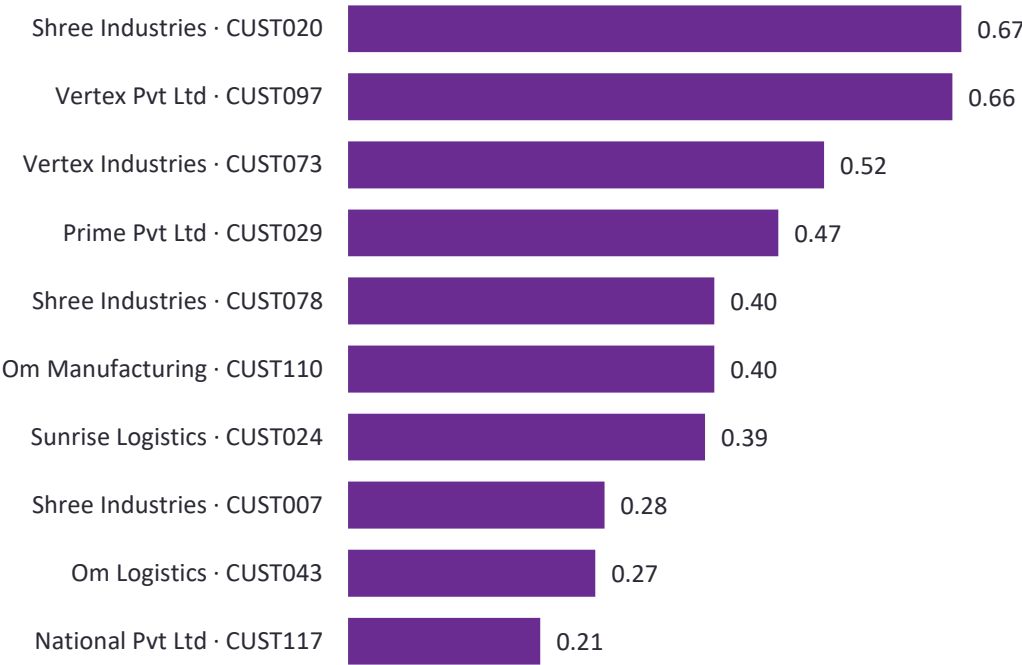
highest state-level sales contribution (Excel dashboard: sales by state and India map)

- **Categories:** Tools & Equipment and Material Handling together ≈ 72% of sales.
- **Regions:** South ≈ 36% of sales; Central is the smallest market (≈ 8%).
- **Timing:** Nov 2025 = ₹2.40 Cr (+161% MoM), then -61% in Dec — seasonal or one-off large orders?

# One in six customers generates almost half of all sales

Top 10 customers by sales (₹ Cr)

MySQL Q4 · p.21



Sales share by customer size



33.5%

Top 10 customers' share of sales

MySQL Q13 · p.30

49.5%

Top 20 of 120 customers

MySQL Q13 · p.30

5.2%

Largest single customer

MySQL Q13 · p.30

**Business implication:** monitor customer concentration while developing opportunities across all size segments (Medium ≈ 43%, Large ≈ 31%, Small ≈ 26%).

Data note: customer names repeat across IDs (e.g. “Shree Industries” appears as CUST007, CUST020, CUST078) — confirm whether these are separate accounts before reading concentration.

# Rank alone is not enough — judge sellers on revenue, profit and order quality



## Meera

Highest salesperson sales contribution in the analysed synthetic dataset

Source: Excel dashboard — sales by salesperson



### Sales volume

Total sales and order count — who brings in the revenue.

Sales\_INR · orders



### Gross profit and margin

Profit contribution — does the revenue come at a healthy margin?

Gross\_Profit\_INR · margin %



### Average order value

Order quality — large deals or many small ones?

Sales ÷ orders

PROPOSED ACTION

Score sellers on revenue, profit and AOV together (Q14) — a top revenue rank with thin margin deserves a closer look.

# The highest-revenue products earn below-portfolio margins

₹12.74 Cr

Sales

–

₹8.64 Cr

Cost

=

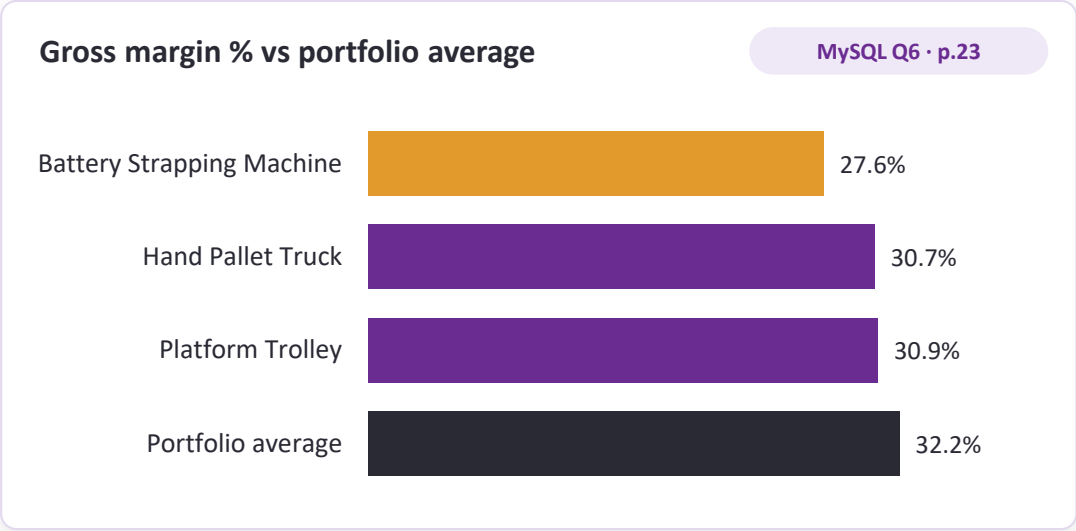
₹4.10 Cr

Gross profit

÷

32.2%

Profit margin



Above-average-revenue products

| Product                   | Sales (₹ Cr) | Share of sales | Margin |
|---------------------------|--------------|----------------|--------|
| Battery Strapping Machine | 4.33         | 34.0%          | 27.6%  |
| Hand Pallet Truck         | 2.76         | 21.7%          | 30.7%  |
| Platform Trolley          | 1.29         | 10.1%          | 30.9%  |

**What-if (illustrative, not a forecast):** lifting Battery Strapping Machine to the 32.2% portfolio margin ≈ +₹19.7 lakh gross profit. These three products = 66% of sales and 60% of gross profit.

FOR MANAGEMENT TO INVESTIGATE

Pricing

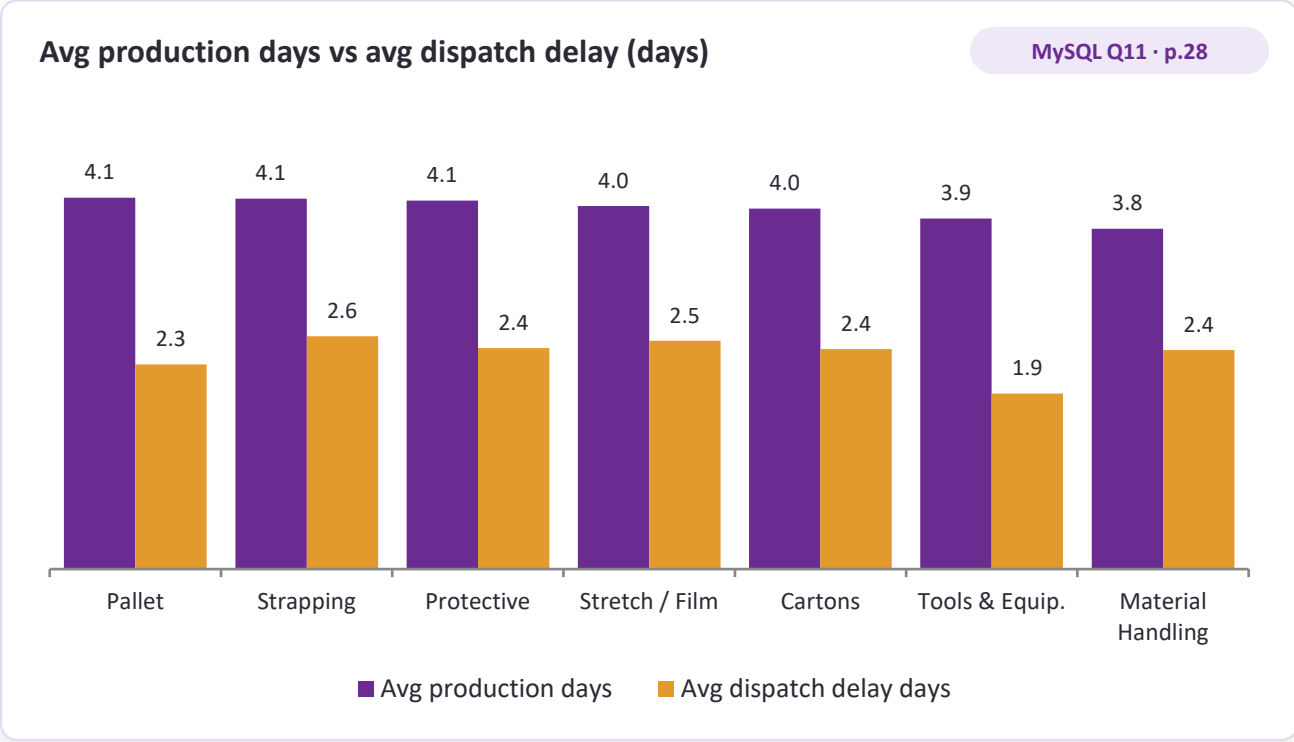
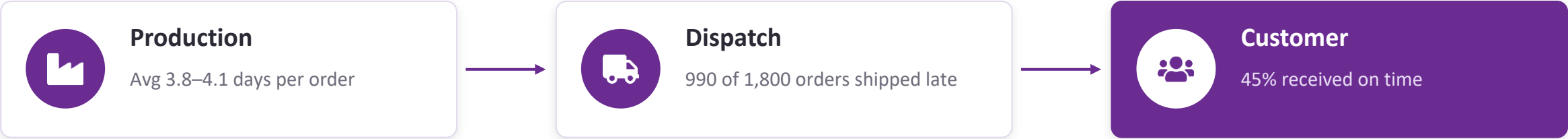
Discounting

Material costs

Product mix

Production costs

# Only 45% of orders ship on time — delays hit every category



**OBSERVED INSIGHT**

- Production time is similar across categories (3.8–4.1 days).
- 51–61% of orders in every category ship late; Strapping is worst (212 of 349).
- Category averages show only a weak link between production time and dispatch delay.

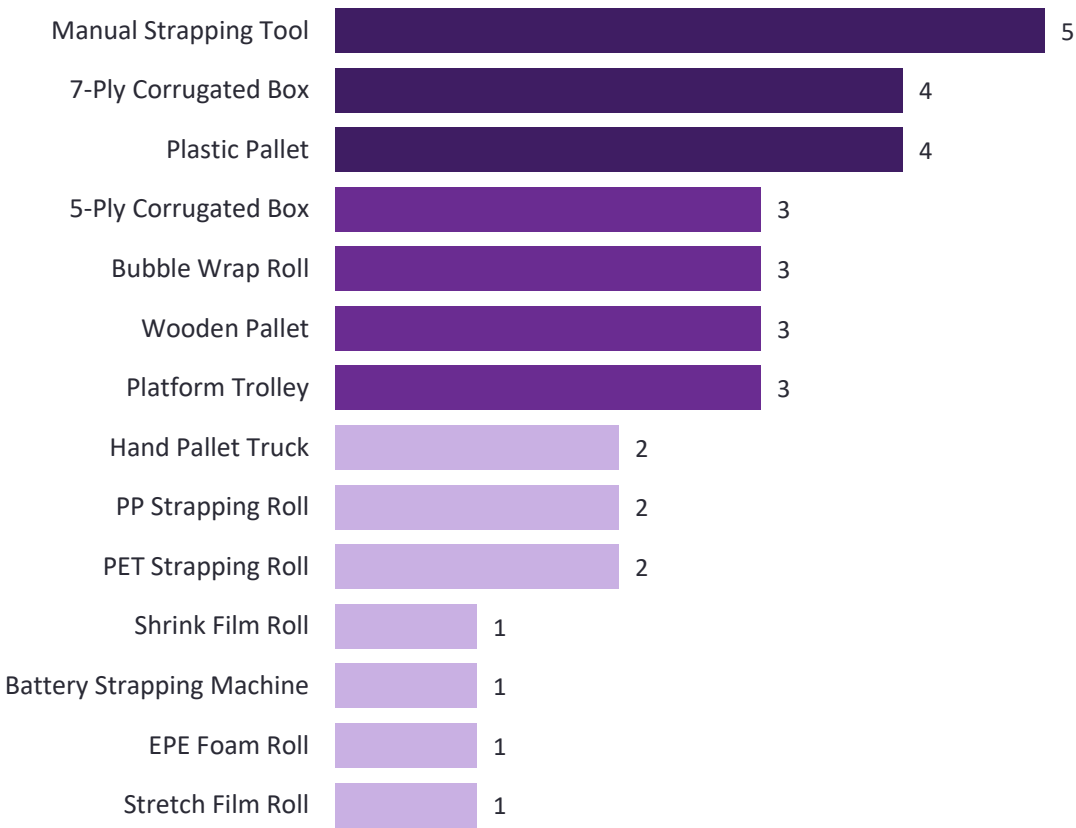
**FOR MANAGEMENT TO INVESTIGATE**

- Is the gap after production (packing, transport, scheduling) the bottleneck?
- Does partial completion (~88% of planned quantity) trigger late dispatch?
- Why does Strapping, the highest-volume category, lag most?

# Every product has dipped below its reorder level at least once

Months flagged “Low Stock” by product

MySQL Q9 · p.26



Latest snapshot (Dec 2025): 7 products low

MySQL Q8 · p.25

| Product              | Closing stock | Reorder level |
|----------------------|---------------|---------------|
| PP Strapping Roll    | 0             | 198           |
| PET Strapping Roll   | 0             | 162           |
| EPE Foam Roll        | 0             | 173           |
| Stretch Film Roll    | 0             | 129           |
| 5-Ply Corrugated Box | 0             | 96            |
| Wooden Pallet        | 11            | 108           |
| Plastic Pallet       | 62            | 179           |

FOR MANAGEMENT TO INVESTIGATE

- Which products repeatedly fall below reorder level? Manual Strapping Tool (5 months).
- Where could stockouts hit fulfilment? Strapping, foam, film and carton lines at zero stock.
- Where is planning weakest? Review reorder levels; also check for excess stock.

# ₹2.16 Cr of sales — 17% — is still uncollected

83.0%

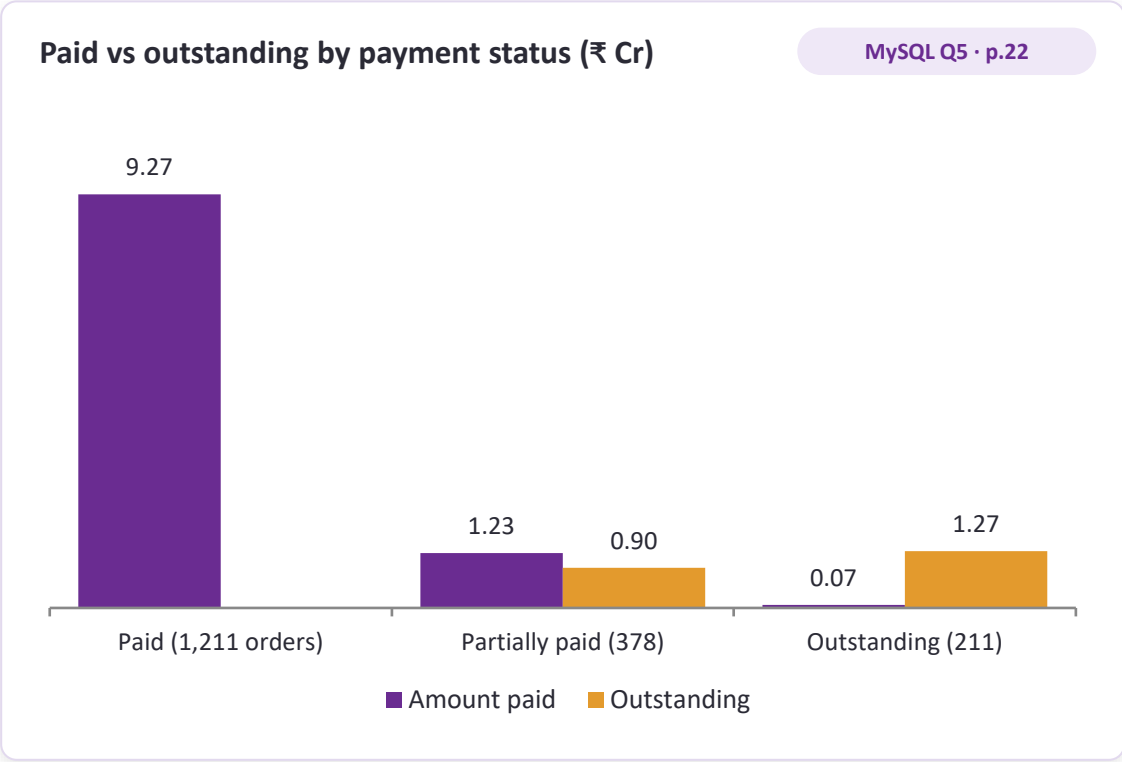
Collection % (paid ÷ sales)

₹2.16 Cr

Outstanding receivables

589 of 1,800

Orders partly paid or unpaid



High sales and high outstanding — top 5 accountsMySQL Q7 · p.24

| Customer                     | Sales (₹ L) | Outstanding (₹ L) | Outst. % |
|------------------------------|-------------|-------------------|----------|
| National Logistics · CUST070 | 19.85       | 19.08             | 96.1%    |
| Shree Enterprises · CUST068  | 15.61       | 9.70              | 62.1%    |
| Shree Industries · CUST007   | 28.16       | 8.93              | 31.7%    |
| United Logistics · CUST072   | 18.52       | 8.66              | 46.8%    |
| Vertex Industries · CUST073  | 52.10       | 7.39              | 14.2%    |

Business implication: sales growth should be monitored together with cash collection. Also review sales vs collection by region (Q15) to find weaker-collection markets.

# Ten observations from the synthetic dataset

1

**₹12.74 Cr sales across 1,800 orders**

Jan–Dec 2025 · synthetic dataset

2

**32.2% weighted gross margin**

₹4.10 Cr gross profit on ₹12.74 Cr sales

3

**45% on-time delivery**

990 of 1,800 orders shipped late — investigate

4

**72% of sales from two categories**

Tools &amp; Equipment and Material Handling

5

**South is the strongest region**

≈ 36% of sales (₹4.63 Cr)

6

**Tamil Nadu leads state-level sales**

Highest state contribution (Excel dashboard)

7

**Medium customers ≈ 43% of sales**

Large ≈ 31% · Small ≈ 26%

8

**Top 20 of 120 customers ≈ 49% of sales**

Top 10 ≈ 33.5% · largest customer 5.2%

9

**Meera leads salesperson sales**

Compare on profit and order value too

10

**₹2.16 Cr outstanding receivables**

83% collected — watch working capital

# Eight proposed actions to turn insight into decisions



01

**Review high-sales, lower-margin products**

Start with Battery Strapping Machine, Hand Pallet Truck and Platform Trolley: price, discount, cost.



02

**Investigate production-to-dispatch bottlenecks**

Trace late orders end to end; begin with Strapping Solutions.



03

**Make on-time delivery a core KPI**

Track weekly by category against promised dates.



04

**Set recurring low-stock alerts**

Closing stock vs reorder level, critical products first.



05

**Monitor customer concentration**

Track top-10 and top-20 share; grow mid-tier accounts.



06

**Prioritise receivable follow-ups**

High-sales, high-outstanding accounts first (e.g. CUST070, CUST068).



07

**Score salespeople on revenue and profit**

Compare sales, gross profit and order value together.



08

**Establish a recurring MIS cadence**

One KPI dictionary, a monthly pack and named owners.

Proposed for management consideration — not claims about practices the company follows today.

# Six predictive opportunities the existing data could support

No forecast values are shown. These are future analytics opportunities — not completed predictions or model outputs.



## Sales forecasting

**Could predict:** Monthly sales forecast from historical orders.

**Built on:** Orders\_Fact · Sales\_INR by month

*Watch-out: only one year of history (2025).*



## Demand forecasting

**Could predict:** Product-level demand from quantity and consumption.

**Built on:** Orders\_Fact · Inventory\_Monthly



## Inventory risk prediction

**Could predict:** Products likely to fall below reorder level.

**Built on:** Closing stock · consumption · reorder level



## Delivery risk prediction

**Could predict:** Orders or categories likely to see dispatch delay.

**Built on:** Production days · promised vs dispatch dates



## Customer risk analysis

**Could predict:** Customers with rising outstanding balances.

**Built on:** Finance · payment status · outstanding



## Profitability forecasting

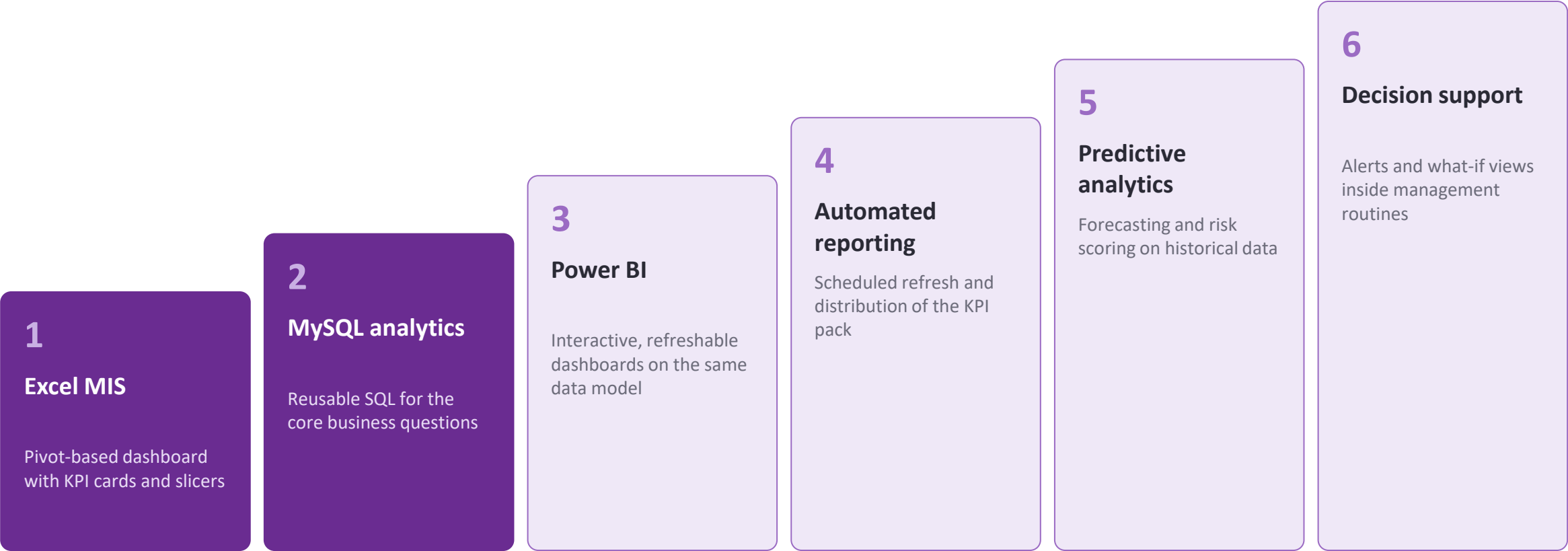
**Could predict:** Category-level revenue and margin trends.

**Built on:** Sales · cost · gross profit by category

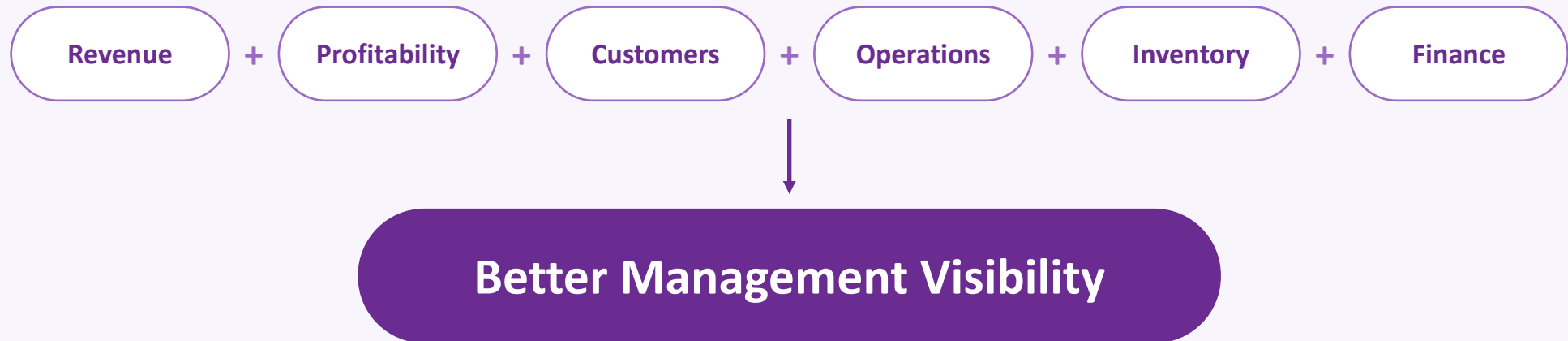
# From static reporting to a recurring decision-support system

 Built in this project       Proposed next

Maturity →



# Integrated business data can move MIS reporting from static reporting to proactive decision support.



APPENDIX

# SQL Evidence

MySQL queries and their outputs behind the findings in this deck — shown as evidence, not as the story.

Screenshots taken from MySQL on the synthetic dataset. Numbers are illustrative only.

|     |                                        |      |
|-----|----------------------------------------|------|
| Q1  | Total revenue, cost and gross profit   | p.19 |
| Q2  | Category-wise sales                    | p.19 |
| Q3  | Region-wise sales                      | p.20 |
| Q4  | Top 10 customers                       | p.21 |
| Q5  | Payment status and outstanding         | p.22 |
| Q6  | High-sales, low-margin products        | p.23 |
| Q7  | High-sales, high-outstanding customers | p.24 |
| Q8  | Products currently low on stock        | p.25 |
| Q9  | Recurring inventory risk               | p.26 |
| Q10 | Production time by category            | p.27 |
| Q11 | Production vs dispatch delays          | p.28 |
| Q12 | Month-over-month sales growth          | p.29 |
| Q13 | Customer revenue concentration         | p.30 |

# Baseline financials and category sales

## Q1 What is the total revenue, cost and gross profit?

### SQL QUERY

```
-- 1. What is the total revenue, total cost and gross profit?  
SELECT  
    round(SUM(Sales_INR),2) AS Total_Sales,  
    round(SUM(Cost_INR),2) AS Total_Cost,  
    round(SUM(Gross_Profit_INR),2) AS Gross_Profit  
FROM orders_fact;
```

### OUTPUT

|   | Total_Sales | Total_Cost  | Gross_Profit |
|---|-------------|-------------|--------------|
| ▶ | 127385268.9 | 86424547.39 | 40960721.51  |

## Q2 Which product categories generate the highest sales?

### SQL QUERY

```
-- 2. Which product categories generate the highest sales?  
▶ SELECT  
    Category,  
    round(SUM(Sales_INR),2) AS Total_Sales,  
    round(SUM(Quantity),2) AS Total_Quantity  
FROM orders_fact  
GROUP BY Category  
ORDER BY Total_Sales DESC;
```

### OUTPUT

| Category                    | Total_Sales | Total_Quantity |
|-----------------------------|-------------|----------------|
| Packaging Tools & Equipment | 51502375.57 | 3630           |
| Material Handling           | 40507679.52 | 3079           |
| Pallet Solutions            | 9886923.78  | 6675           |
| Protective Packaging        | 9122067.72  | 8317           |
| Strapping Solutions         | 7999467.09  | 75505          |
| Stretch / Film Solutions    | 7652416.06  | 10883          |
| Customized Carton Boxes     | 714339.16   | 9731           |

## Region-wise sales

### SQL QUERY

```
-- 3. 3. Which regions generate the most sales?
SELECT
    Region,
    COUNT(DISTINCT Order_ID) AS Total_Orders,
    round(SUM(Sales_INR),2) AS Total_Sales,
    round(SUM(Gross_Profit_INR),2) AS Gross_Profit
FROM orders_fact
GROUP BY Region
ORDER BY Total_Sales DESC;
```

### OUTPUT

| Region  | Total_Orders | Total_Sales | Gross_Profit |
|---------|--------------|-------------|--------------|
| South   | 628          | 46250719.54 | 14825034.35  |
| North   | 530          | 28453585.06 | 9564582.47   |
| East    | 241          | 23114895.84 | 7346850.75   |
| West    | 235          | 19926103.8  | 5920951.53   |
| Central | 166          | 9639964.66  | 3303302.41   |

### READ-OUT

South leads with 628 orders and ₹4.63 Cr ( $\approx 36\%$  of sales, ₹1.48 Cr gross profit). North ₹2.85 Cr, East ₹2.31 Cr, West ₹1.99 Cr, Central ₹0.96 Cr.

## Top 10 customers by sales

### SQL QUERY

```
-- 4. Who are our top 10 customers?  
SELECT  
    Customer_ID,  
    Customer_Name,  
    COUNT(DISTINCT Order_ID) AS Total_Orders,  
    round(SUM(Sales_INR),2) AS Total_Sales  
FROM orders_fact  
GROUP BY Customer_ID, Customer_Name  
ORDER BY Total_Sales DESC  
LIMIT 10;
```

### OUTPUT

| Customer_ID | Customer_Name     | Total_Orders | Total_Sales |
|-------------|-------------------|--------------|-------------|
| CUST020     | Shree Industries  | 28           | 6674405.11  |
| CUST097     | Vertex Pvt Ltd    | 14           | 6597913.96  |
| CUST073     | Vertex Industries | 15           | 5209820.27  |
| CUST029     | Prime Pvt Ltd     | 10           | 4726689.34  |
| CUST078     | Shree Industries  | 16           | 3970806.59  |
| CUST110     | Om Manufacturing  | 31           | 3969005.16  |
| CUST024     | Sunrise Logistics | 18           | 3932216.22  |
| CUST007     | Shree Industries  | 13           | 2816154.22  |
| CUST043     | Om Logistics      | 17           | 2672100.34  |
| CUST117     | National Pvt Ltd  | 14           | 2120395.53  |

### READ-OUT

CUST020 (Shree Industries) is the largest account at ₹0.67 Cr. Customer names repeat across IDs — confirm these are distinct accounts before reading concentration.

# Payment status and outstanding receivables

## SQL QUERY

```
-- 5. What is the payment status of our orders?  
SELECT  
    Payment_Status,  
    COUNT(DISTINCT Order_ID) AS Total_Orders,  
    round(SUM(Sales_INR),2) AS Total_Sales,  
    round(SUM(Amount_Paid_INR),2) AS Amount_Paid,  
    round(SUM(Outstanding_INR),2) AS Outstanding  
FROM orders_fact  
GROUP BY Payment_Status  
ORDER BY Outstanding DESC;
```

## OUTPUT

|   | Payment_Status | Total_Orders | Total_Sales | Amount_Paid | Outstanding |
|---|----------------|--------------|-------------|-------------|-------------|
| ▶ | Outstanding    | 211          | 13407180.23 | 736354.22   | 12670826.01 |
|   | Partially Paid | 378          | 21249099.4  | 12293168.81 | 8955930.59  |
|   | Paid           | 1211         | 92728989.27 | 92728989.27 | 0           |

## READ-OUT

1,211 orders fully paid (₹9.27 Cr). Partially paid: 378 orders, ₹0.90 Cr outstanding. Outstanding status: 211 orders, ₹1.27 Cr outstanding. Total ₹2.16 Cr = 17% of sales; collection ≈ 83%.

# High-sales products vs profit margin

## SQL QUERY

```
-- 6. Which products generate high sales but low profit margins?
WITH product_summary AS (
  SELECT
    Product_ID,
    Product_Name,
    Category,
    SUM(Sales_INR) AS Product_Sales,
    SUM(Gross_Profit_INR) AS Product_Profit
  FROM orders_fact
  GROUP BY Product_ID, Product_Name, Category
),
avg_sales AS (
  SELECT AVG(Product_Sales) AS avg_product_sales
  FROM product_summary
)
SELECT
  ps.Product_ID,
  ps.Product_Name,
  ps.Category,
  ROUND(ps.Product_Sales, 2) AS Total_Sales,
  ROUND(ps.Product_Profit, 2) AS Gross_Profit,
  ROUND(ps.Product_Profit / NULLIF(ps.Product_Sales, 0) * 100, 2) AS Profit_Margin_Percent
FROM product_summary ps
CROSS JOIN avg_sales a
WHERE ps.Product_Sales > a.avg_product_sales
ORDER BY Profit_Margin_Percent;
```

## OUTPUT

|   | Product_ID | Product_Name              | Category                    | Total_Sales | Gross_Profit | Profit_Margin_Percent |
|---|------------|---------------------------|-----------------------------|-------------|--------------|-----------------------|
| ▶ | P010       | Battery Strapping Machine | Packaging Tools & Equipment | 43255011.04 | 11943655.2   | 27.61                 |
|   | P014       | Hand Pallet Truck         | Material Handling           | 27596256.98 | 8482398.56   | 30.74                 |
|   | P013       | Platform Trolley          | Material Handling           | 12911422.54 | 3992193.08   | 30.92                 |

## READ-OUT

Products with above-average revenue: Battery Strapping Machine ₹4.33 Cr at 27.61%, Hand Pallet Truck ₹2.76 Cr at 30.74%, Platform Trolley ₹1.29 Cr at 30.92% — all below the 32.2% portfolio margin.

# High-sales customers with high outstanding balances

## SQL QUERY

```
-- 7. Which customers have high sales but also high outstanding payments?
WITH customer_summary AS (
    SELECT
        Customer_ID,
        Customer_Name,
        SUM(Sales_INR) AS Total_Sales,
        SUM(Outstanding_INR) AS Outstanding
    FROM orders_fact
    GROUP BY Customer_ID, Customer_Name
)
SELECT
    Customer_ID,
    Customer_Name,
    Total_Sales,
    Outstanding,
    ROUND(Outstanding / NULLIF(Total_Sales, 0) * 100, 2) AS Outstanding_Percentage
FROM customer_summary
WHERE Total_Sales > (SELECT AVG(Total_Sales) FROM customer_summary)
    AND Outstanding > (
        SELECT AVG(Outstanding)
        FROM customer_summary
    )
ORDER BY Outstanding DESC;
```

## OUTPUT

| Customer_ID | Customer_Name           | Total_Sales        | Outstanding        | Outstanding_Percentage |
|-------------|-------------------------|--------------------|--------------------|------------------------|
| CUST070     | National Logistics      | 1985023.34         | 1908301.68         | 96.13                  |
| CUST068     | Shree Enterprises       | 1561394.21         | 970073.84          | 62.13                  |
| CUST007     | Shree Industries        | 2816154.2199999997 | 892953.07          | 31.71                  |
| CUST072     | United Logistics        | 1851500.92         | 866168.1200000001  | 46.78                  |
| CUST073     | Vertex Industries       | 5209820.27         | 739397.9           | 14.19                  |
| CUST024     | Sunrise Logistics       | 3932216.2199999997 | 663562.82          | 16.88                  |
| CUST076     | Metro Manufacturing     | 1413306.06         | 477092.05          | 33.76                  |
| CUST043     | Om Logistics            | 2672100.3400000003 | 449546.38999999996 | 16.82                  |
| CUST016     | Shree Pvt Ltd           | 1136444.23         | 444704.35          | 39.13                  |
| CUST020     | Shree Industries        | 6674405.11         | 440308.81000000006 | 6.6                    |
| CUST078     | Shree Industries        | 3970806.59         | 440126.54000000004 | 11.08                  |
| CUST042     | National Manufacturi... | 1446552.0699999998 | 437761.4           | 30.26                  |
| CUST110     | Om Manufacturing        | 3969005.16         | 425861.3           | 10.73                  |
| CUST036     | National Industries     | 1063182.0500000003 | 390697.5           | 36.75                  |
| CUST083     | Metro Industries        | 1697511.32         | 387378.93          | 22.82                  |
| CUST071     | Prime Manufacturing     | 1410482.7299999997 | 358434.01          | 25.41                  |
| CUST029     | Prime Pvt Ltd           | 4726689.3400000001 | 313118.28          | 6.62                   |
| CUST092     | Metro Industries        | 1147974.9699999997 | 310277.79          | 27.03                  |
| CUST001     | Prime Industries        | 1154164.0399999998 | 306034.10000000003 | 26.52                  |
| CUST069     | United Logistics        | 1206996.77         | 201758.11          | 16.72                  |
| CUST014     | National Industries     | 1430252.07         | 201587.92          | 14.09                  |

## READ-OUT

Customers above average on both sales and outstanding. CUST070 has 96% of its sales outstanding; CUST068 has 62%.

# Products currently in low-stock status

SQL QUERY

```
-- 8. Which products currently have low inventory?
SELECT
    Product_ID,
    Product_Name,
    Category,
    Snapshot_Month,
    Closing_Stock,
    Reorder_Level,
    Stock_Status
FROM inventory_monthly
WHERE Stock_Status = 'Low Stock'
ORDER BY Snapshot_Month DESC, Closing_Stock;
```

OUTPUT

| Product_ID | Product_Name            | Category                    | Snapshot_Month | Closing_Stock | Reorder_Level | Stock_Status |
|------------|-------------------------|-----------------------------|----------------|---------------|---------------|--------------|
| P011       | 5-Ply Corrugated Box    | Customized Carton Boxes     | 2025-12-01     | 0             | 96            | Low Stock    |
| P005       | Stretch Film Roll       | Stretch / Film Solutions    | 2025-12-01     | 0             | 129           | Low Stock    |
| P004       | EPE Foam Roll           | Protective Packaging        | 2025-12-01     | 0             | 173           | Low Stock    |
| P002       | PET Strapping Roll      | Strapping Solutions         | 2025-12-01     | 0             | 162           | Low Stock    |
| P001       | PP Strapping Roll       | Strapping Solutions         | 2025-12-01     | 0             | 198           | Low Stock    |
| P007       | Wooden Pallet           | Pallet Solutions            | 2025-12-01     | 11            | 108           | Low Stock    |
| P008       | Plastic Pallet          | Pallet Solutions            | 2025-12-01     | 62            | 179           | Low Stock    |
| P012       | 7-Ply Corrugated Box    | Customized Carton Boxes     | 2025-11-01     | 0             | 158           | Low Stock    |
| P009       | Manual Strapping Tool   | Packaging Tools & Equipment | 2025-10-01     | 59            | 155           | Low Stock    |
| P014       | Hand Pallet Truck       | Material Handling           | 2025-09-01     | 0             | 176           | Low Stock    |
| P009       | Manual Strapping Tool   | Packaging Tools & Equipment | 2025-09-01     | 0             | 199           | Low Stock    |
| P007       | Wooden Pallet           | Pallet Solutions            | 2025-09-01     | 0             | 179           | Low Stock    |
| P002       | PET Strapping Roll      | Strapping Solutions         | 2025-09-01     | 0             | 187           | Low Stock    |
| P006       | Shrink Film Roll        | Stretch / Film Solutions    | 2025-08-01     | 14            | 68            | Low Stock    |
| P013       | Platform Trolley        | Material Handling           | 2025-08-01     | 58            | 200           | Low Stock    |
| P010       | Battery Strapping Ma... | Packaging Tools & Equipment | 2025-08-01     | 101           | 108           | Low Stock    |
| P008       | Plastic Pallet          | Pallet Solutions            | 2025-08-01     | 151           | 177           | Low Stock    |
| P003       | Bubble Wrap Roll        | Protective Packaging        | 2025-07-01     | 27            | 166           | Low Stock    |
| P013       | Platform Trolley        | Material Handling           | 2025-06-01     | 0             | 198           | Low Stock    |
| P009       | Manual Strapping Tool   | Packaging Tools & Equipment | 2025-06-01     | 0             | 168           | Low Stock    |
| P001       | PP Strapping Roll       | Strapping Solutions         | 2025-06-01     | 12            | 98            | Low Stock    |

READ-OUT

Latest snapshot shown (Dec 2025): 7 products flagged Low Stock, 5 of them with zero closing stock.

# Products that repeatedly fall below reorder level

SQL QUERY

```
-- 9. Which products repeatedly fall below the reorder level?
SELECT
  Product_ID,
  Product_Name,
  COUNT(*) AS Low_Stock_Months
FROM inventory_monthly
WHERE Stock_Status = 'Low Stock'
GROUP BY Product_ID, Product_Name
ORDER BY Low_Stock_Months DESC;
```

READ-OUT

All 14 products were flagged at least once. Manual Strapping Tool leads with 5 low-stock months; 7-Ply Corrugated Box and Plastic Pallet follow with 4 each.

OUTPUT

| Product_ID | Product_Name            | Low_Stock_Months |
|------------|-------------------------|------------------|
| P009       | Manual Strapping Tool   | 5                |
| P012       | 7-Ply Corrugated Box    | 4                |
| P008       | Plastic Pallet          | 4                |
| P011       | 5-Ply Corrugated Box    | 3                |
| P003       | Bubble Wrap Roll        | 3                |
| P007       | Wooden Pallet           | 3                |
| P013       | Platform Trolley        | 3                |
| P014       | Hand Pallet Truck       | 2                |
| P001       | PP Strapping Roll       | 2                |
| P002       | PET Strapping Roll      | 2                |
| P006       | Shrink Film Roll        | 1                |
| P010       | Battery Strapping Ma... | 1                |
| P004       | EPE Foam Roll           | 1                |
| P005       | Stretch Film Roll       | 1                |

# Production time by category

## SQL QUERY

```
-- 10. Which product categories have the longest production time?
SELECT
    Category,
    COUNT(DISTINCT Order_ID) AS Total_Orders,
    ROUND(AVG(Production_Days), 2) AS Avg_Production_Days,
    ROUND(
        AVG('Production_Completion_%') * 100,
        2
    ) AS Avg_Production_Completion_Percent
FROM production
GROUP BY Category
ORDER BY Avg_Production_Days DESC;
```

## OUTPUT

| Category                    | Total_Orders | Avg_Production_Days | Avg_Production_Completion_Percent |
|-----------------------------|--------------|---------------------|-----------------------------------|
| Pallet Solutions            | 227          | 4.10                | 87.96                             |
| Strapping Solutions         | 349          | 4.09                | 87.27                             |
| Protective Packaging        | 288          | 4.07                | 89.36                             |
| Stretch / Film Solutions    | 326          | 4.01                | 89.68                             |
| Customized Carton Boxes     | 373          | 3.98                | 87.17                             |
| Packaging Tools & Equipment | 121          | 3.87                | 89.52                             |
| Material Handling           | 116          | 3.76                | 88.58                             |

## READ-OUT

Average production time sits in a narrow 3.76–4.10 day band; completion averages 87–90% of planned quantity.

# Production delays vs dispatch delays

## SQL QUERY

```
-- 11. Are production delays related to dispatch delays?
SELECT
  o.Category,
  COUNT(DISTINCT o.Order_ID) AS Total_Orders,

  ROUND(AVG(p.Production_Days), 2)
    AS Avg_Production_Days,

  ROUND(AVG(o.Dispatch_Delay_Days), 2)
    AS Avg_Dispatch_Delay_Days,

  SUM(
    CASE
      WHEN o.Dispatch_Delay_Days > 0
      THEN 1
      ELSE 0
    END
  ) AS Delayed_Orders

FROM orders_fact o
JOIN production p
  ON o.Order_ID = p.Order_ID

GROUP BY o.Category
ORDER BY Avg_Dispatch_Delay_Days DESC;
```

## OUTPUT

| Category                    | Total_Orders | Avg_Production_Days | Avg_Dispatch_Delay_Days | Delayed_Orders |
|-----------------------------|--------------|---------------------|-------------------------|----------------|
| Strapping Solutions         | 349          | 4.09                | 2.57                    | 212            |
| Stretch / Film Solutions    | 326          | 4.01                | 2.52                    | 186            |
| Protective Packaging        | 288          | 4.07                | 2.44                    | 150            |
| Customized Carton Boxes     | 373          | 3.98                | 2.43                    | 198            |
| Material Handling           | 116          | 3.76                | 2.42                    | 59             |
| Pallet Solutions            | 227          | 4.10                | 2.26                    | 121            |
| Packaging Tools & Equipment | 121          | 3.87                | 1.94                    | 64             |

## READ-OUT

Late dispatch affects 990 of 1,800 orders (51–61% per category). Strapping Solutions is worst (2.57 days average delay). Across the seven categories, production time and dispatch delay show only a weak link.

# Month-over-month sales growth

SQL QUERY

```
-- 12. What is the month-over-month sales growth?
WITH monthly_sales AS (
  SELECT
    DATE_FORMAT(Order_Date, '%Y-%m') AS Sales_Month,
    SUM(Sales_INR) AS Total_Sales
  FROM orders_fact
  GROUP BY DATE_FORMAT(Order_Date, '%Y-%m')
),

monthly_comparison AS (
  SELECT
    Sales_Month,
    Total_Sales,
    LAG(Total_Sales) OVER (
      ORDER BY Sales_Month
    ) AS Previous_Month_Sales
  FROM monthly_sales
)

SELECT
  Sales_Month,
  Total_Sales,
  Previous_Month_Sales,
  ROUND(
    (Total_Sales - Previous_Month_Sales)
    / NULLIF(Previous_Month_Sales, 0) * 100,
    2
  ) AS MoM_Growth_Percent
FROM monthly_comparison
ORDER BY Sales_Month;
```

OUTPUT

|   | Sales_Month | Total_Sales        | Previous_Month_Sales | MoM_Growth_Percent |
|---|-------------|--------------------|----------------------|--------------------|
| ▶ | 2025-01     | 12334588.069999999 | NULL                 | NULL               |
|   | 2025-02     | 7624707.270000001  | 12334588.069999999   | -38.18             |
|   | 2025-03     | 7566207.99         | 7624707.270000001    | -0.77              |
|   | 2025-04     | 7429268.430000001  | 7566207.99           | -1.81              |
|   | 2025-05     | 9495502.870000001  | 7429268.430000001    | 27.81              |
|   | 2025-06     | 10690254.029999996 | 9495502.870000001    | 12.58              |
|   | 2025-07     | 11443425.710000006 | 10690254.029999996   | 7.05               |
|   | 2025-08     | 8401747.230000002  | 11443425.710000006   | -26.58             |
|   | 2025-09     | 9761272.660000002  | 8401747.230000002    | 16.18              |
|   | 2025-10     | 9203782.189999994  | 9761272.660000002    | -5.71              |
|   | 2025-11     | 24015791.83        | 9203782.189999994    | 160.93             |
|   | 2025-12     | 9418720.620000003  | 24015791.83          | -60.78             |

READ-OUT

Nov 2025 sales of ₹2.40 Cr are +160.9% vs Oct, followed by -60.8% in Dec.

# Customer revenue concentration

## SQL QUERY

```
-- 13. Which customers contribute to the majority of company revenue?
WITH customer_sales AS (
    SELECT
        Customer_ID,
        Customer_Name,
        SUM(Sales_INR) AS Total_Sales
    FROM orders_fact
    GROUP BY Customer_ID, Customer_Name
),
ranked_customers AS (
    SELECT
        Customer_ID,
        Customer_Name,
        Total_Sales,
        SUM(Total_Sales) OVER (ORDER BY Total_Sales DESC) AS Cumulative_Sales,
        SUM(Total_Sales) OVER () AS Company_Sales
    FROM customer_sales
)
SELECT
    Customer_ID,
    Customer_Name,
    Total_Sales,
    ROUND(Total_Sales / Company_Sales * 100,2) AS Sales_Percentage,
    ROUND(Cumulative_Sales / Company_Sales * 100,2) AS Cumulative_Sales_Percentage
FROM ranked_customers
ORDER BY Total_Sales DESC;
```

## OUTPUT

| Customer_ID | Customer_Name        | Total_Sales        | Sales_Percentage | Cumulative_Sales_Percentage |
|-------------|----------------------|--------------------|------------------|-----------------------------|
| CUST020     | Shree Industries     | 6674405.11         | 5.24             | 5.24                        |
| CUST097     | Vertex Pvt Ltd       | 6597913.9600000001 | 5.18             | 10.42                       |
| CUST073     | Vertex Industries    | 5209820.27         | 4.09             | 14.51                       |
| CUST029     | Prime Pvt Ltd        | 4726689.3400000001 | 3.71             | 18.22                       |
| CUST078     | Shree Industries     | 3970806.59         | 3.12             | 21.34                       |
| CUST110     | Om Manufacturing     | 3969005.16         | 3.12             | 24.45                       |
| CUST024     | Sunrise Logistics    | 3932216.2199999997 | 3.09             | 27.54                       |
| CUST007     | Shree Industries     | 2816154.2199999997 | 2.21             | 29.75                       |
| CUST043     | Om Logistics         | 2672100.3400000003 | 2.1              | 31.85                       |
| CUST117     | National Pvt Ltd     | 2120395.5300000003 | 1.66             | 33.51                       |
| CUST070     | National Logistics   | 1985023.34         | 1.56             | 35.07                       |
| CUST032     | Vertex Industries    | 1876150.2300000004 | 1.47             | 36.54                       |
| CUST072     | United Logistics     | 1851500.92         | 1.45             | 38                          |
| CUST066     | Om Logistics         | 1755871.1600000004 | 1.38             | 39.38                       |
| CUST022     | United Manufactu...  | 1740223.8699999999 | 1.37             | 40.74                       |
| CUST083     | Metro Industries     | 1697511.32         | 1.33             | 42.07                       |
| CUST091     | Metro Pvt Ltd        | 1621069.5900000003 | 1.27             | 43.35                       |
| CUST080     | Vertex Industries    | 1582442.58         | 1.24             | 44.59                       |
| CUST047     | Global Pvt Ltd       | 1574711.69         | 1.24             | 45.82                       |
| CUST068     | Shree Enterprises    | 1561394.21         | 1.23             | 47.05                       |
| CUST111     | National Manufact... | 1556428.92         | 1.22             | 48.27                       |
| CUST002     | Global Pvt Ltd       | 1524530.8          | 1.2              | 49.47                       |
| CUST043     | National Manufact... | 1446553.8600000008 | 1.14             | 50.6                        |

## READ-OUT

Top 10 customers = 33.5% of sales; top 20 = 49.5%. The largest single customer is 5.2%.